25-FYP-307



AI Driven Virtual Makeup Try-on

DEPARTMENT OF COMPUTER SCIENCE

Department of Computer Science

NATIONAL TEXTILE UNIVERSITY

Description

The AI-Based Virtual Makeup Try-On application is a desktop system that enables users to apply virtual makeup in real-time using facial recognition and augmented reality. Developed with Python, the application supports multiple cosmetic products including foundation, lipstick, eyeliner, eyeshadow, and eyebrow coloring. It uses facial landmark tracking to overlay makeup accurately on the user's live camera feed. With a modern GUI, offline functionality, and adjustable intensity features, the system offers a hygienic, interactive, and user-friendly experience for personal use, salons, and online cosmetic previews bridging beauty with intelligent, real-time digital innovation.

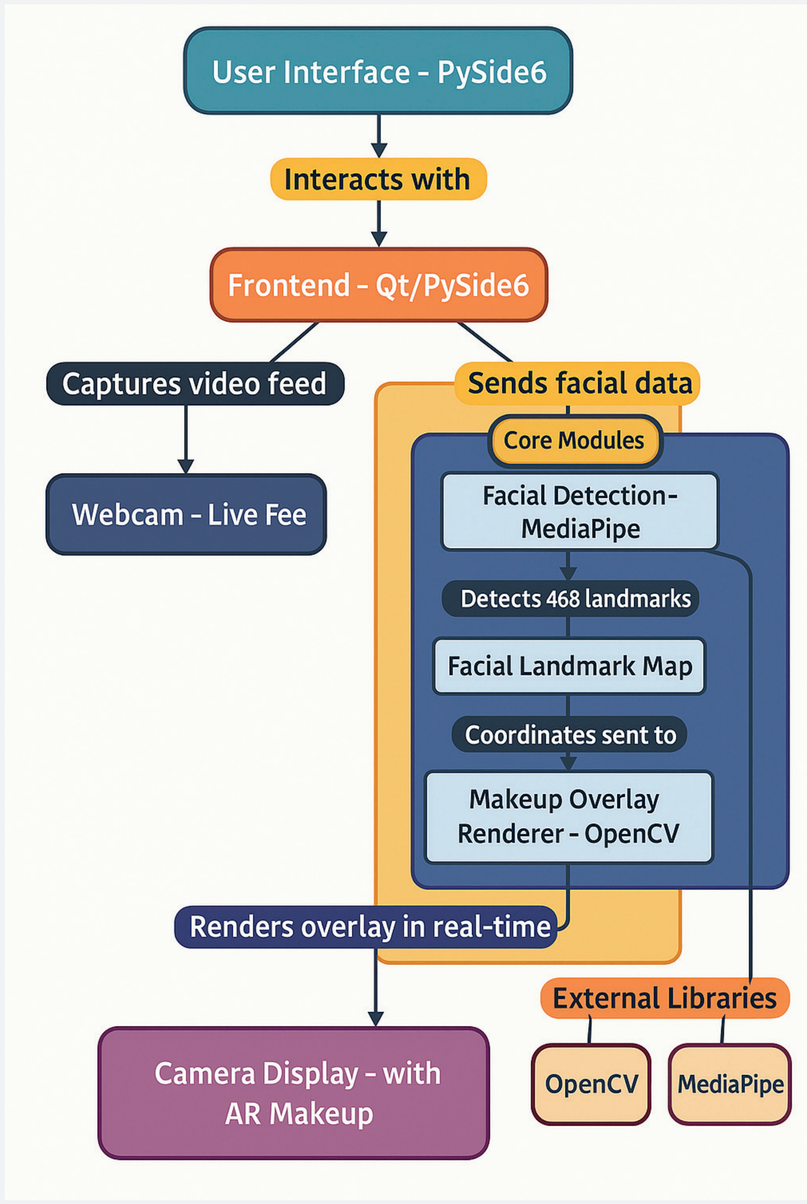
Main Features

- Real-Time Virtual Makeup Application
- Facial Landmark Detection & Augmented Reality
- Adjustable Makeup Intensity & Effects
- Offline Functionality for Desktop Use
- Interactive & User-Friendly Interface

Tools & Technologies

- Python
- Tensorflow
- PyQt6
- ONNX Runtime
- CUDA Acceleration

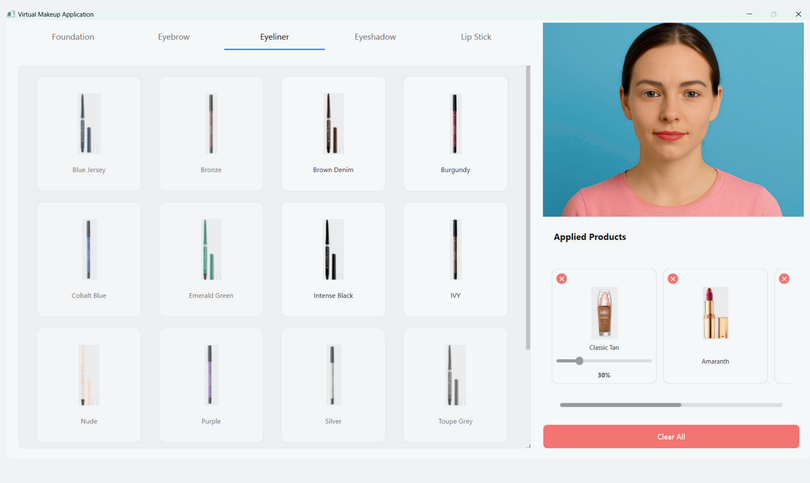
Process Flow Diagram

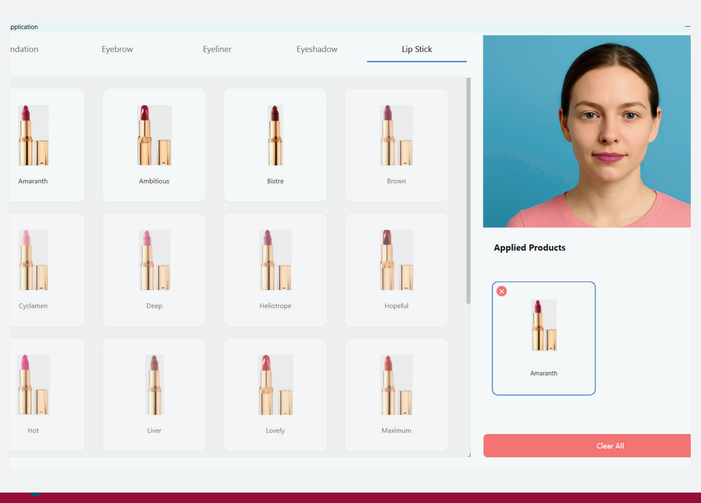


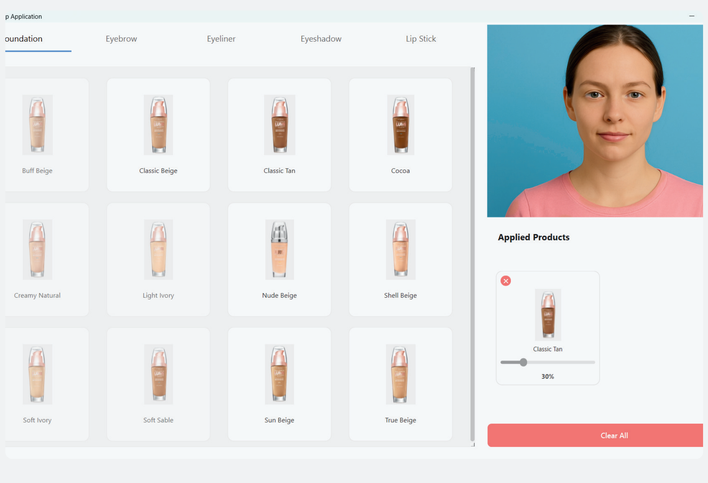
Target Market

- Individuals seeking virtual cosmetic trials at home
- Beauty salons and makeup artists
- Cosmetic retailers and e-commerce platforms
- Researchers and developers in AR & beauty tech

Screenshots







Group Members

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FINAL YEAR PROJECT DISPLAY 2025

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